DevOps Case Study Task for Kubernetes, Terraform, Ansible, and Grafana

Objective:

In this case study, the candidate is required to demonstrate their skills in setting up infrastructure using **Terraform**, configuring the environment using **Ansible**, deploying a containerized application to **Kubernetes**, and implementing basic **monitoring** using **Grafana**. The task should be designed to take around **2 hours** to complete.

Scenario:

Your organization has requested to deploy a sample web application in a Kubernetes cluster. This application should be monitored, and infrastructure as code (IaC) should be used to manage the setup.

Your task is to:

1. **Provision infrastructure** using **Terraform**.
2. **Deploy an application** in a Kubernetes cluster.
3. **Configure monitoring** for the application using **Grafana**.
4. **Automate the setup** using **Ansible**.

Task Breakdown:

1. **Infrastructure Setup using Terraform (30 minutes)**
    - Use Terraform to provision a Kubernetes cluster locally (use **Minikube**, **K3s**, or **Kind** for simplicity).
    - The cluster should have at least:
        - **1 control plane node**.
        - **1 worker node**.
    - Configure necessary networking (use basic networking configurations or Calico/CNI).

2. **Application Deployment in Kubernetes (30 minutes)**
    - Create a simple **Nginx** or **Node.js** containerized web application (you can use Docker Hub images for simplicity).
    - Deploy the application using **Kubernetes manifests** or **Helm** charts.
    - Create the following Kubernetes objects:
        - **Deployment** with 2 replicas.
        - **Service** to expose the application internally.
        - **ConfigMap** for storing some basic environment variables (e.g., `APP_NAME=webapp`).

            - Optionally, create an **Ingress** rule to route traffic if needed.
3. **Monitoring Setup with Grafana (40 minutes)**
    - Install and configure **Prometheus** and **Grafana** in the Kubernetes cluster.
    - Use a basic **Prometheus exporter** (e.g., **Node Exporter** or **Kube State Metrics**) to monitor the health of the Kubernetes nodes or the web application.
    - Create a **Grafana dashboard** to monitor the following:
        - Pod CPU and memory usage.
        - Application uptime.
    - Ensure that Grafana is accessible from outside the cluster using a Kubernetes **service** or **Ingress**.

4. **Automation with Ansible (20 minutes) - optional**
    - Write an **Ansible playbook** to automate the provisioning and deployment of:
        - Kubernetes cluster setup using Terraform.
        - Deployment of the application to the Kubernetes cluster.
        - Installation and configuration of Prometheus and Grafana.

Prerequisites for the Candidate:

- Installed tools: Terraform, Ansible, Kubernetes (Minikube/Kind/K3s), Docker, and Helm.